

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Dr. Aqsa Aslam and Mr. Shoaib Raza	
Student Roll No:	Section No:

Instructions:

- Return the question paper along with the answer script.
- All questions must be answered in the answer script and according to the sequence given in the question paper. There are 9 questions on 3 pages.**
- In case of any ambiguity, you may make an assumption, however, your assumption should not contradict any statement in the question paper.

Time: 180 minutes

Max Points: 100 points

Question 1:

[5x3=15 points]

Final Year students were giving a demo of their e-commerce Website to their supervisor. Their Web Page has 10 different objects. Supervisor asked the following questions during VIVA VOICE.

- a) How many RRT will be required to download those objects if they have used the Persistent and Non-Persistent version of HTTP?

Solution:

20 RTT in case of Non Persistent and 11 RTT in case of Persistent RTT.

- b) What is Head of Line (HOL) blocking in the HTTP/2 version and how it was mitigated?

Solution:

HTTP1.1: introduced multiple, pipelined GETs over single TCP connection

- server responds *in-order* (FCFS: first-come-first-served scheduling) to GET requests
- with FCFS, small object may have to wait for transmission (head-of-line (HOL) blocking) behind large object(s)
- loss recovery (retransmitting lost TCP segments) stalls object transmission

- c) How HTTP/3 has increased the performance of Web browsers?

Solution:

HTTP/3: adds security, per object error- and congestion-control (more pipelining) over UDP.

- moving transport-layer functions to application layer, on top of UDP
 - HTTP/3: QUIC

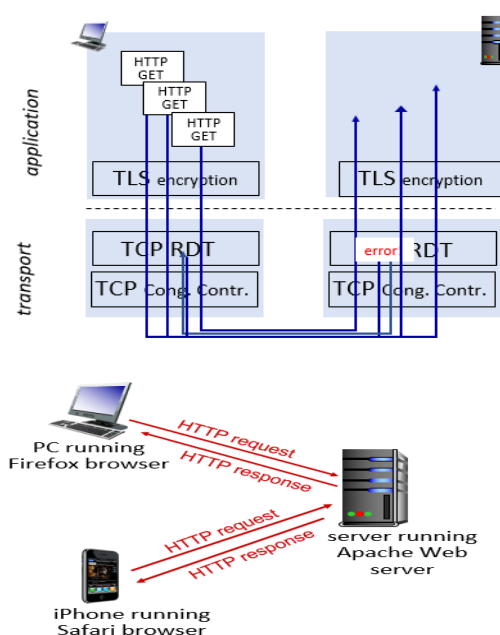
- d) Is HTTP a client-server or a peer-to-peer paradigm? Explain your answer.

Solution:

Client/server model:

Client: browser that requests, receives, (using HTTP protocol) and “displays” Web objects

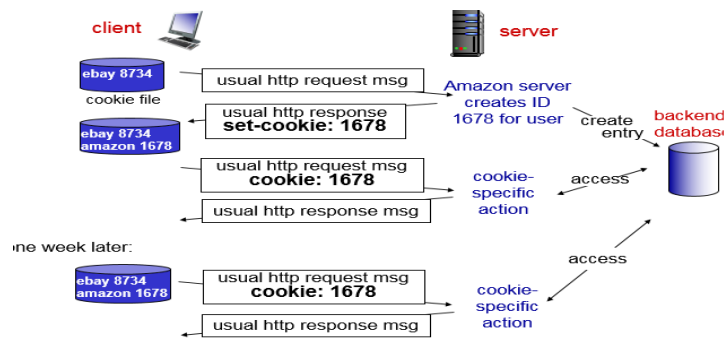
Server: Web server sends (using HTTP protocol) objects in response to requests



- e) How this e-commerce website can keep browsing trends of its customers?

Solution:

Cookies keeps the state of the customer session as shown below.



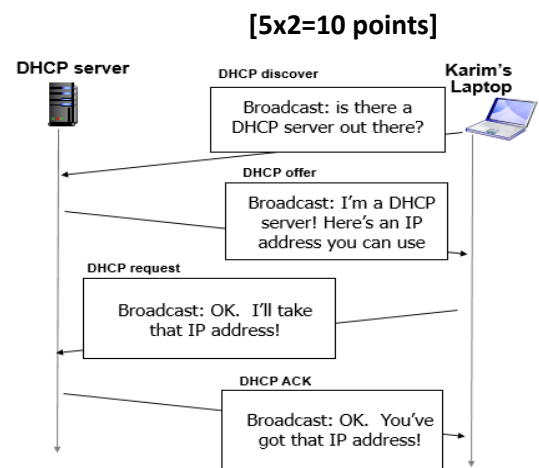
Question 2:

Suppose a user Karim enters the university with his laptop. He is in room S2 in the CS block. Assume that the IT department of the University is running various services to support network operations e.g. DHCP, DNS, File storage, and web application (e.g. Flex). Describe in steps how Karim's laptop:

- a) Connects to the Internet.

Solution:

- Connect to WiFi and sets IP address manually by asking around (10% marks only).
- Connects to especially setup-up WiFi connection. Set OS to get IP address using DHCP (50% marks if there are missing exchanges).



- b) Initiates a query to flex.nu.edu.pk.

Solution:

When karim initiates a query to flex.nu.edu.pk, the host makes DNS query, query is sent to its local DNS server which has local cache of recent name-to-address translation pairs. If not query will be forwarded to TLD Server and Root level servers.

OR

- Do a DNS lookup to get IP address of flex domain.
- Show all HTTP exchanges to get the default home page.

- c) Connects to another student Talha sitting in the same room (S2).

Solution:

He can connect to Talha by using ARP OR RARP protocol.

OR

- Both you and Talha connected to LAN: Need a client-server application (Local web-server) or P2P application (Bit-Torrent client) running on Talha machine.
- Both you and Talha not connect to LAN: Make a Bluetooth, WiFi direct peer to peer (P2P) connection and use custom applications (such as smart wristwatch sync etc. and file transfer)
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- d) Connects to another instructor Farhan in the EE department.

Solution:

- Cannot connect to Farhan using Bluetooth or WiFi direct as he is too far away.
- If Farhan is connect to LAN same as part (c) above option (i).

- e) Suppose accessing the File storage has become a bottleneck due to multiple users trying to access the hosted CN video lectures. What application-level solution will you propose to overcome this problem?

Solution:

This is similar to the file upload to multiple peer we studied in the class. Client-server model needs more computer, storage and network bandwidth as the number of users accessing the shared file folders grows to a large number (video lecture where size is ~1000s of MB).

A P2P based application on each peer (like Bit torrent running on LAN) will allow a new user to use multiple copies of lectures those stored on different peers (as well as server copy) to take part of the download. Each peer PC will provide the bandwidth hence removing the bottleneck.

Question 3:

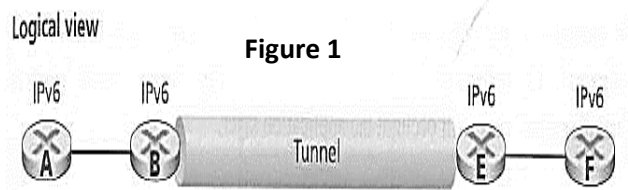
[2x5=10 points]

- a) Why IP fragmentation was implemented in IPv4? Explain how IPv6 solves the issue for which IP fragmentation was designed in IPv4.

Solution:

- Each IP datagram needs to fit in a frame at the source PC using its MTU size (e.g. ~1400). If it encounters a MTU size of < 1400 while passing through Internet (a mesh connection of routers) the router with lower MTU interface fragments the IP datagram into two or more datagrams. These datagrams travel the Internet without change (if they haven't encountered another lower MTU) and are assembled back by the destination network layer.
- IPv6 performs a MTU path discovery to the destination IP address before sending the packets and uses the lowest MTU.

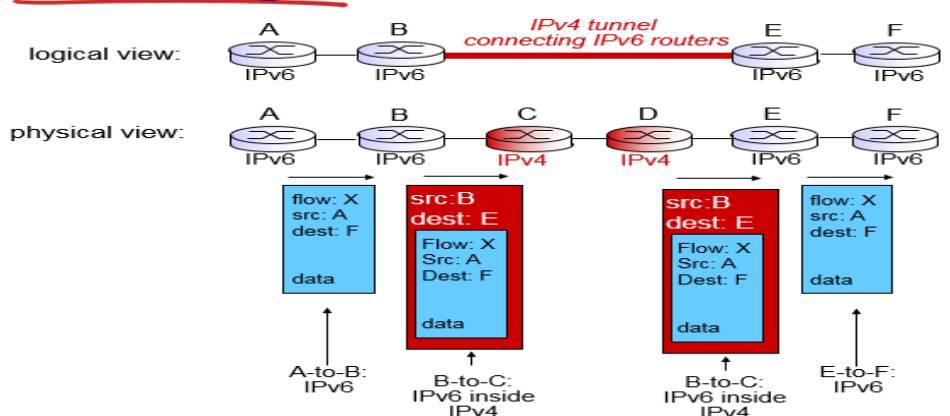
- b) ISPs are slowly changing their IP networks to IPv6. This means they have to interconnect between groups of routers running IPv4 with other groups running IPv6. Suppose a scenario shown in figure 1, where IPv6 packets need to traverse an IPv4 tunnel. How is this tunnel practically implemented?



Solution:

It is implemented using tunneling. IPv6 datagram carried as payload in IPv4 datagram among IPv4 routers as shown below.

Tunneling



Question 4:

[2x5=10 points]

- a) TCP connection has currently estimated RTT of 30 ms with a deviation of 3 ms. What is the value of the retransmission timer after the next acknowledgement comes in after 28 ms?

Solution:

$$\text{EstimatedRTT} = (1-\alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

$$= 0.875 \times 30 \text{ ms} + 0.125 \times 28 \text{ ms} = 29.75 \text{ ms}$$

$$\text{DevRTT} = (1-\beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{EstimatedRTT}|$$

$$= 0.75 \times 3 \text{ ms} + 0.25 \times |28 \text{ ms} - 29.75 \text{ ms}| = 2.6875 \text{ ms}$$

$$\text{RTO} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$

$$= 29.75 \text{ ms} + 4 \times 2.6875 \text{ ms} = 40.5 \text{ ms}$$

- b) Does UDP require a mechanism to estimate the RTT between sender and receiver? Briefly explain your answer.

Solution:

No, UDP does not perform retransmissions and, thus, does not require an RTT-based timeout timer.

Question 5:

[2x5=10 points]

Consider figure 2, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link. Suppose that the packet length is $L = 8000$ bits and that the link transmission rate along the link to the router on the right is $R = 100$ Mbps.

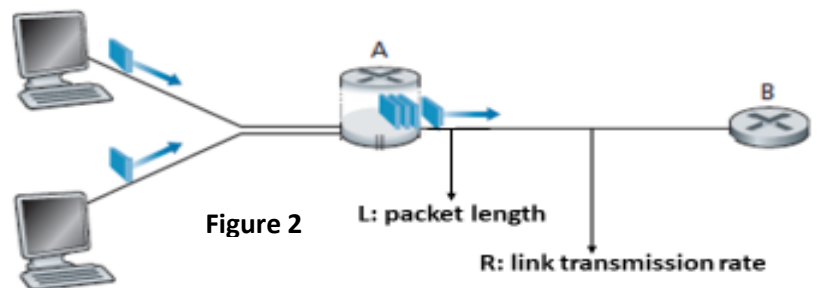


Figure 2

- a) What is the transmission delay?

Solution:

The transmission delay = $L/R = 8000 \text{ bits} / 100,000,000 \text{ bps}$

Transmission delay = 0.00008 seconds

- b) Assume the average rate is 30. What will be traffic intensity?

Solution:

Traffic intensity (I) = $(aL)/R = (30 \cdot 8000) / (100,000,000)$

Traffic intensity (I) = 0.0024

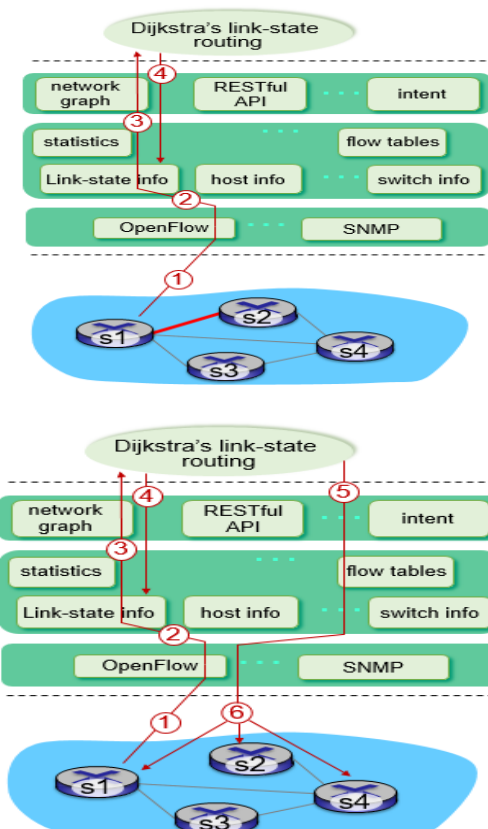
Question 6:

[2x5 =10 points]

Software-defined networking technology is an approach to network management that enables dynamic and programmatically efficient network. Now answer the following questions:

- a) What are four key characteristics of an SDN architecture?

Solution:



- ① S1, experiencing link failure uses OpenFlow port status message to notify controller
- ② SDN controller receives OpenFlow message, updates link status info
- ③ Dijkstra's routing algorithm application has previously registered to be called when ever link status changes. It is called.
- ④ Dijkstra's routing algorithm access network graph info, link state info in controller, computes new routes

- ⑤ link state routing app interacts with flow-table-computation component in SDN controller, which computes new flow tables needed
- ⑥ controller uses OpenFlow to install new tables in switches that need updating

- b) How SDN controller makes use of different network applications to provide link state routing in the SND setup? Explain it in steps.

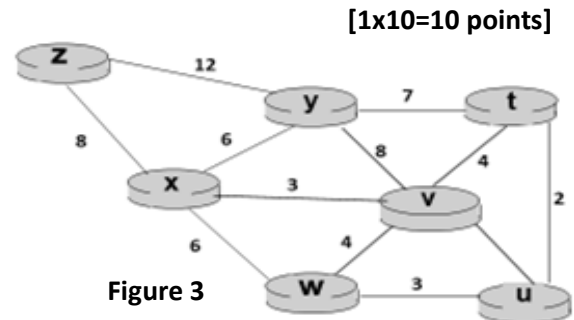
Solution:

1. Flow-based forwarding.
2. Separation of data plane and control plane.
3. Network control functions:
4. A programmable network.

Question 7:

Consider the network shown in figure 3. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from w to all network nodes. Show how the algorithm works by computing a table. State of Initial routing table is shown below

Solution:



Steps	Node (N')	D(x), p(x)	D(u), p(u)	D(v), p(v)	D(t), p(t)	D(y), p(y)	D(z), p(z)
0	w	6,w	3,w	4,w	∞	∞	∞
1	wu	6,w		4,w	5,u	∞	∞
2	wuv	6,w			5,u	12,v	∞
3	wuvt	6,w				12,v	∞
4	wuvtx					12,v	14,x
5	wuvtxy						14,x
6	wuvtxyz						

Question 8:

You have been given the IP address 192.168.9.0/24. Now you have to assign IP addresses to the different department's LAN as shown in network figure 4. The network has the following addressing requirements.

- a) The Department-1 LAN-1 will require 50 host IP addresses.

Determine the following.

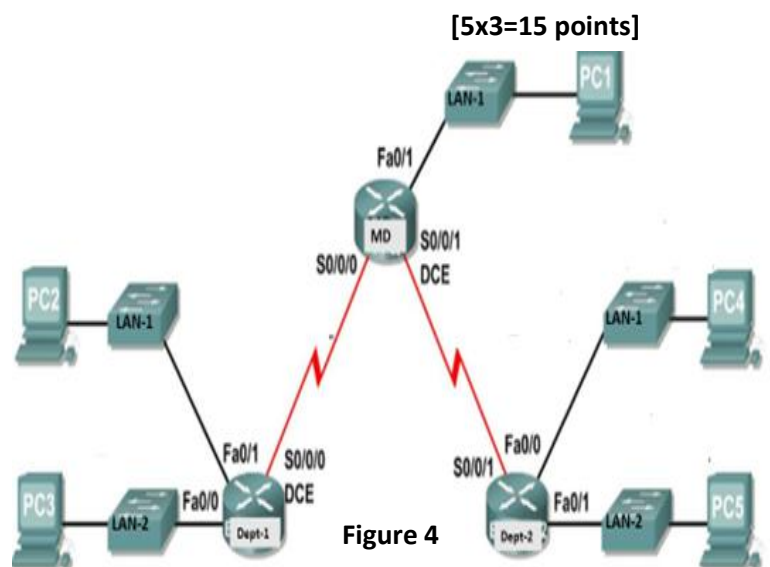
- Number of Bit Borrowed= 2
- Total Number of Subnets= $2^2 = 4$
- Total Number of Host Addresses= 64
- Number of Usable Addresses=62
- Custom Subnet Mask= 255.255.255.192
- Subnet Range(Network & Broadcast address)

192.168.9.0 ---192.168.9.63

192.168.9.128---192.168.9.191

192.168.9.64 ---192.168.9.127

192.168.9.192 ---192.168.9.255



b) The Department-1 LAN-2 will require 120 host IP addresses.

Determine the following.

- Number of Bit Borrowed= 1
- Total Number of Subnets= $2^1 = 2$
- Total Number of Host Addresses= 128
- Number of Usable Addresses= 126
- Custom Subnet Mask= 255.255.255.128
- Subnet Range(Network & Broadcast address)
192.168.9.0 ---192.168.9.127 192.168.9.128 ---192.168.9.255

c) The Department-2 LAN 1will require 10 host IP addresses.

Determine the following.

- Number of Bit Borrowed= 4
- Total Number of Subnets= $2^4 = 16$
- Total Number of Host Addresses= 16
- Number of Usable Addresses= 14
- Custom Subnet Mask= 255.255.255.240
- Subnet Range(Network & Broadcast address)
192.168.9.0 ---192.168.9.15 192.168.9.16 ---192.168.9.31
192.168.9.32---192.168.9.47 192.168.9.48 ---192.168.9.63
192.168.9.64 ---192.168.9.79 192.168.9.80 ---192.168.9.95
192.168.9.96 ---192.168.9.111 192.168.9.112 ---192.168.9.127
192.168.9.128 ---192.168.9.143 192.168.9.144 ---192.168.9.159
192.168.9.160 ---192.168.9.175 192.168.9.176 ---192.168.9.191
192.168.9.192 ---192.168.9.207 192.168.9.208 ---192.168.9.223
192.168.9.224 ---192.168.9.239 192.168.9.240 ---192.168.9.255

d) The Department-2 LAN 2will require 18 host IP addresses.

Determine the following.

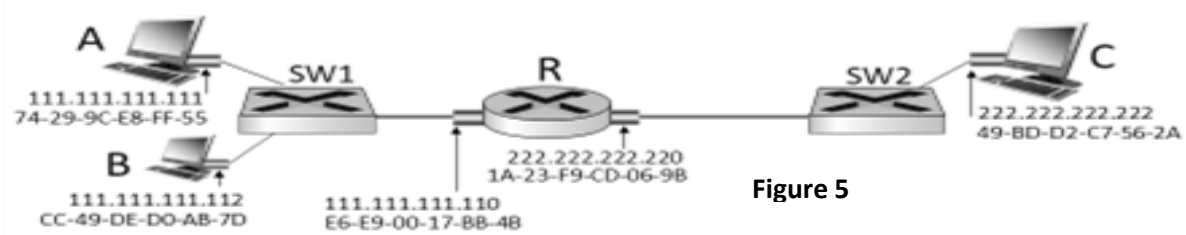
- Number of Bit Borrowed= 3
- Total Number of Subnets= $2^3 = 8$
- Total Number of Host Addresses= 32
- Number of Usable Addresses= 30
- Custom Subnet Mask= 255.255.255.224
- Subnet Range(Network & Broadcast address)
192.168.9.0 ---192.168.9.31 192.168.9.32 ---192.168.9.63
192.168.9.64---192.168.9.95 192.168.9.96 ---192.168.9.127
192.168.9.128 ---192.168.9.159 192.168.9.160 ---192.168.9.191
192.168.9.192 ---192.168.9.223 192.168.9.224---192.168.9.255

e) The MD LAN-1 will require 28 host IP addresses.

We can assign any of the remaining subnet from part (d).

Question 9:**[3+3+4=10 points]**

Consider figure 5 and answer the following questions:



- a) Consider sending an IP datagram from PC-A to PC-B. PC- A issues an ARP request because it needs to send a packet to PC- B. In this scenario, what will happen next?

Solution:

PC-B will send an ARP reply with its MAC address (CC-49-DE-D0-AB-7D).

- b) Suppose PC- A would like to send IP datagram to PC - C on a remote LAN, however there are currently no mappings in the ARP cache. How will the device obtain a destination MAC address?

Solution:

It will send an ARP request for the MAC address of the default gateway (R) i-e (E6-E9-00-17-BB-4B)

- c) What will be the content of the destination IP-address field in the header of the IP datagram sent by PC- A to PC-C?

Solution:

The IP address of PC-C (222.222.222.222)

Best of Luck!